

Gauss gun — Solution

X25 (2025) — English translation

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-2.00 pts

Attention! If your steel ball rolls under someone else's table, then it is forbidden to take it out! Spare balls and magnets are not issued! Marking ANY kind of mark on an aluminum gutter will result in a penalty point. Losing ANY equipment other than balls or a magnet will result in a penalty point. Intentionally changing the angle of an aluminum gutter will disqualify you.

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A1

0.10 pts

Specify the value h . Determine the installation angle α .

Let's measure the height of the lower horizontal part of the gutter $h = 24.9 \text{ cm}$. Let's measure the height of the top point $h + \Delta h = 50.7 \text{ cm}$. In this case, the length of the long part of the gutter is $L = 80.0 \text{ cm}$. Using the fact that $\sin \alpha = \Delta h / L$ we find the angle α

Answer: \textbf{Answer:}

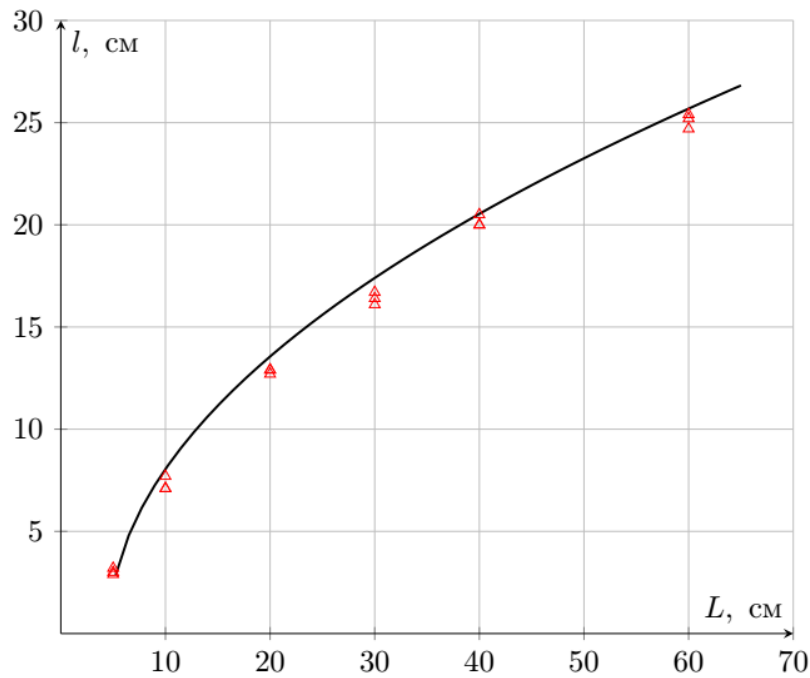
$$h = 24.9 \text{ cm} \quad \alpha = 18.8^\circ$$

A2

1.00 pts

Measure the dependence of the flight range l on the length of the inclined section of the ball path L in the range from 0 to 70 cm for 5 values L . Repeat each measurement at least 3 times. All measurements l must be taken using ball drop marks on A3 sheets. A3 sheets are required to be handed in at the end of the work.

Answer: \textbf{Answer:} $L, \text{ cm } l_1, \text{ cm } l_2, \text{ cm } l_3, \text{ cm } \langle l \rangle, \text{ cm}$ 10.0 7.1 7.1 7.7 7.8 20.0 12.7 12.9 12.9 13.3 30.0 16.1 16.4 16.7 16.9 40.0 20.0 20.0 20.5 20.7 60.0 24.7 25.2 25.4 25.6



A3

0.40 pts

Obtain a theoretical expression relating the quantities l and L . Assume that the ball moves along the chute all the time without slipping; the angle between the generatrices of the chute is straight. Express your answer using h, α, γ, l, L

Kinetic energy of the ball taking into account the rotational energy:

$$K = \frac{Mv^2}{2} + \frac{I\omega^2}{2},$$

where $I = \frac{2}{5}MR^2$. The speed of the ball points that touch the corner must be zero, so $\omega R/\sqrt{2} = v$. As a result, on the inclined section of the chute, the ESE has the form

$$K = \Delta\Pi \Rightarrow \frac{9Mv^2}{10} = MgL \sin \alpha$$

After passing the corner of the chute, γK of energy is lost, so the ball flies out of the corner with a horizontal speed u such that

$$u^2 = \frac{10}{9}(1 - \gamma)gL \sin \alpha$$

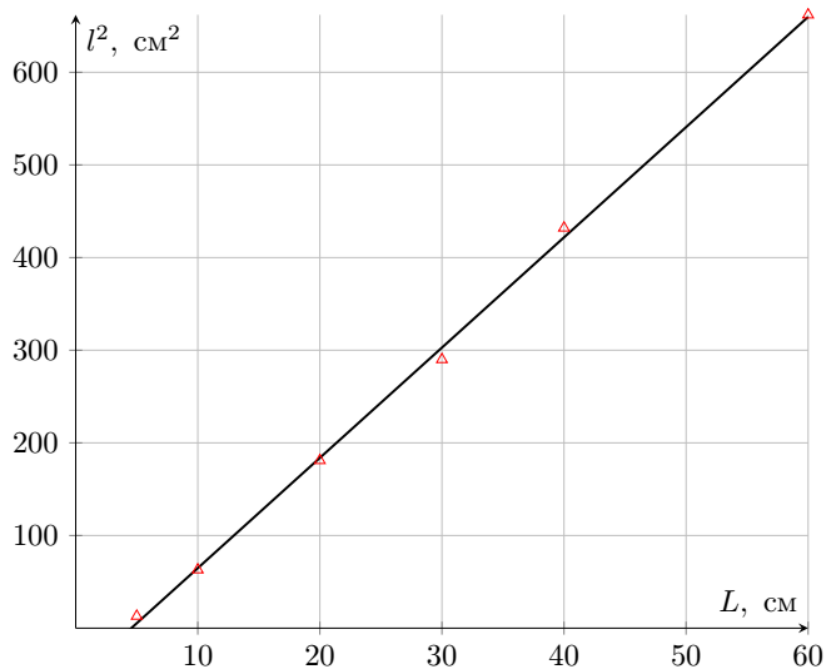
The ball, under the influence of gravity acceleration, overcomes the height h in time $t = \sqrt{2h/g}$, and therefore

$$l^2 = u^2 t^2 = \frac{20}{9}(1 - \gamma)Lh \sin \alpha$$

A4

0.60 pts

Using the result of the previous paragraph, construct a graph of l versus L in coordinates in which it will be linear.



A5

0.20 pts

Find the slope coefficient of the graph constructed in the previous paragraph. Determine the loss factor γ .

The graph in axes l^2 from L turns out to be linear with a slope coefficient

$$k_{\mathbf{A4}} = \frac{20}{9}(1 - \gamma)h \sin \alpha = 11.9 \text{ cm}$$

The coefficient γ can vary quite a lot from corner to corner.

Answer: \textbf{Answer:}

$$\gamma = 1 - \frac{9k_{\mathbf{A4}}}{20h \sin \alpha} = 0.36$$

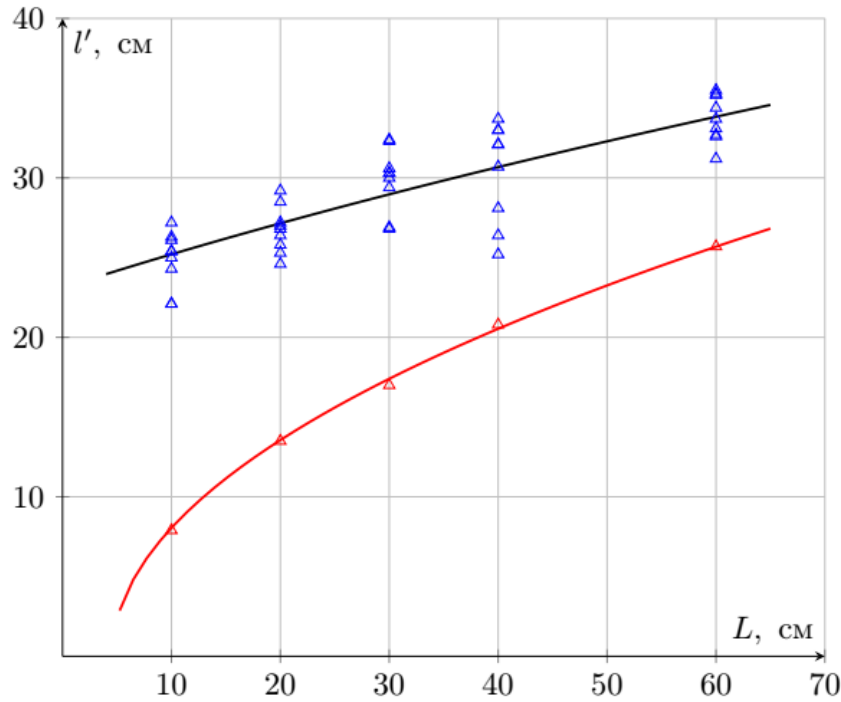
B1

2.00 pts

Measure the dependence of the flight range l' of the emitted ball on the length of the inclined section of the path of the incoming ball L . Take measurements for the same values of quantity L that you used in Part A. For each value of L , take at least 5 measurements of quantity l' . All measurements l must be taken using marks from the fall of the ball on A3 sheets.

All measurements l must be taken using marks from the fall of the ball on A3 sheets.

Answer: \textbf{Answer:} $L, \text{ cm}$ $\langle l' \rangle, \text{ cm}$ 10.0 25.4 20.0 26.8 30.0 30.3 40.0 30.5 60.0 34.0



B2

0.30 pts

Obtain an expression for the dependence $E(l)$.

From the kinematics described above, we remove the rotational motion

$$E = \frac{Mu^2}{2} = \frac{Mgl^2}{4h}$$

Answer: \textbf{Answer:}

$$E = \frac{Mgl^2}{4h}$$

B3

0.40 pts

Using the results of steps A2 and B1, calculate the addition ΔE to the kinetic energy of the projectile translational motion obtained during interaction with the gun, for each experimental value. Calculate the energy using the formula obtained in the previous paragraph, discarding constant factors in front of the function of l .

$$\Delta E = l'^2 - l^2$$

Answer: \textbf{Answer:} L, cm $\Delta E, cm^2$ $\Delta E, mJ$ 10.0 584 2.36 20.0 534 2.16 30.0 627 2.53 40.0 497 2.00 60.0 492 1.99

B4

0.40 pts

Plot a graph of ΔE versus l in coordinates in which it is linear.

Energy ratio:

$$E' = (E + W_1)(1 - \beta) + W_2,$$

therefore

$$\Delta E = -\beta l^2 + W_1(1 - \beta) + W_2$$

and the graph must be plotted in coordinates ΔE from l^2